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1964-2014



Unit – II

Environmental Pollution and its Control

Department of Applied Sciences, BVCOE, New Delhi



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ENVIRONMENTAL POLLUTION

- **Pollution** : Any undesirable change in the physical, chemical or biological characteristics of any component of the environment which cause harmful effects on various forms of life.
- **Pollutant**: any substance if present in an undesirable concentration in the wrong place at the wrong time. Eg. Fertilizers, carbon dioxide etc.



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CLASSIFICATION OF POLLUTION

1. AIR POLLUTION
2. WATER POLLUTION
3. SOIL POLLUTION
4. SOLID WASTE POLLUTION
5. **MARINE POLLUTION**
6. NOISE POLLUTION
7. THERMAL POLLUTION
8. NUCLEAR/Radiation POLLUTION

SOIL POLLUTION

- forms the uppermost layer of the land, is the **most precious natural resource** because it supports the whole life system
- provides food and fodder in the form of vegetation

Due to **urbanization and industrialization**, this valuable source is getting polluted.

Soil pollution is defined as “**undesirable characteristics in the physicochemical properties, which have adverse effects on human beings, plant growth and animals**”.



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CAUSES AND SOURCES/TYPES OF SOIL POLLUTION

The major sources or causes of soil pollution includes:

- **Mining and quarrying,**
- **household waste,**
- **industrial waste,**
- **biomedical waste,**
- **fertilizers and pesticides**

Various chemical substances present in them reach the soil and eventually in all water bodies , through rainfall, irrigation, drainage etc. Thus, polluting and disturbing the natural balance of the ecosystem.



Some important pollution sources are:

- 1. Domestic Waste-** Wastes from the residences are usually found dumped on soil include **polythene, garbage, organic waste, paper, glass, plastics, clothes, metal containers** etc. Plastic metal and glass do not degrade easily and cause more harm to the soils. Human and Animal excretion add pathogens to soil.
- 2. Industrial Waste-** Wastes from **textiles, tannery, chemicals industries, electroplating, glass, distilleries, paper, petroleum, thermal power plants and cement industries**, etc enter the soil as it is disposed off or dumped as such. These wastes contain inorganic and organic materials that alter and change natural composition and chemistry of soil.

- **FERTILIZERS** - The excessive use of nitrogenous fertilizers leads to nitrate discharge in the soil and ultimately pollute ground water.

When human being drink this nitrate rich water the nitrates are converted to nitrites by the action of intestinal flora. The nitrites have an affinity to hemoglobin and combines with it to form **methaemoglobin which interferes the** oxygen carrying capacity of blood. This phenomenon is called **methomoglobinaemia**. It is common in the infants and often results into **blue baby syndrome**. Its symptoms include blue coloration of the skin, vascular and respiratory problems, headache, etc..



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4. Pesticides- Pesticides not only kill the target organisms but also affect large variety of living things including human beings.

Example - DDT (Dichloro Diphenyl Trichloroethane) is a persistent type of pesticide, do not break down easily and tend to accumulate in the soil and water. It may accumulate in the bodies of animals and over a period of time their concentration increases if the animal is unable to flush them out of its body. Thus leading to the phenomena called **bio-accumulation**. When an affected animal is further eaten by another carnivore these pesticides are concentrated in the body of consumer. This phenomenon of acquiring increasing levels of substances in the bodies of higher tropic level organism is called **Bio magnification/Bioaccumulation**.



- **Radioactive material**

The radioactive wastes are generally released into the soil from

- nuclear explosion,
- atmosphere fallout from nuclear dusts,
- discharges from radioactive laboratories.

Radioactive elements such as **Thorium, Uranium, Radium, and Cesium** are commonly found in the environment that keeps emitting radiations.



EFFECTS ON LIVING ORGANISMS:

- **Non-degradable substances like glass, plastic, polythene and metal** containers do not degrade easily and cause toxicity to the soil for very long periods.
- They also **alter the physico-chemical properties of soil and disturb their natural composition**, e.g. blue baby syndrome in infants (Rajasthan and Punjab)
- Radiation emitted by radioactive substance dumped on soil persist for a long time. These radiations are known to bio-accumulate in plants also. Radiation can enter human beings when such exposed plants and food containing them are ingested resulting in mutations and genetic disorders.



SOIL POLLUTION CONTROL

Soil pollution can be **reduced or controlled** by practicing the following steps:

1. Effluents should be **properly treated** before discharging them into soil.
2. Solid wastes should be **properly collected and disposed off** by appropriate methods.
3. From the domestic or industrial waste **recovery of useful materials** should be done.
4. By practicing the “**4 R rule**” i.e. **Refuse, Reduce, Recover and Recycle**, the soil pollution can be reduced very efficiently.
5. Biodegradable organic waste should be used for **composting**

THERMAL POLLUTION

It is defined as the **presence of waste heat in the water** which can cause undesirable changes in the natural environment.

The **discharge of warm water into a water body** is usually called a thermal pollution.

Sources

- It occurs when an industry removes water from a source, uses the water for cooling purposes and then returns the heated water to its source.
- Power plants heat water to convert it into steam, to drive the turbines that generate electricity. For efficient functioning of the steam turbines, the steam is condensed into water after it leaves the turbines. This condensation is done by taking water from a water body to absorb the heat. This heated water, which is at least 15°C higher than the normal is discharged back into the water body.
- Refineries, steel mills, nuclear power plants

Effects of Thermal pollution

- **DO content of water decreases** as the solubility of oxygen in water reduces at high temperature.
- **Toxicity of pesticides, detergents etc. in the effluents increases** with rise in temperature
- Many species undergo **thermal shock**, and **temperature tolerant species take over** existing flora and fauna.
- **Fish migration is affected** due to formation of various thermal zones
- **Metabolic activities of aquatic organisms increase** at high temperature and require more oxygen, thereby reducing DO levels

Control of thermal pollution

- **Cooling ponds** – Water from condensers is stored in ponds where natural evaporation cools the water which can then be recirculated or discharged in nearby water body
- **Spray ponds** – the water from condensers is received in spray ponds where water is sprayed through nozzles where fine droplet's heat is dissipated to atmosphere.
- **Cooling towers** – a) **Wet cooling towers** – Hot water is sprayed over baffles, cool air entering from sides takes away the heat and cools the water. This cool water can then be discharged or recycled.
- b) **Dry cooling towers** – the heated water flows in a system of pipes. Air is passed over these hot pipes with fans. There is no water loss in this method but installation and operation cost is higher.

Noise pollution

- Noise may not seem as harmful as the contamination of air or water but it is a pollution problem that affects human health and can contribute to a general deterioration of environmental quality.
- **Noise is any undesirable and unwanted sound.** Not all sound is noise. It is not a substance that can accumulate in the environment like most other pollutants.

Sound is measured in a unit called the 'Decibel'.

- **Decibel levels of common sounds**

• dB	Environmental Condition	60	Conversational speech (at 1m)
• 0	Threshold of hearing	70	Average radio
• 10	Rustle of leaves	74	Light traffic noise
• 40	Library	90	Subway train
• 50	Quiet office	100	Symphony orchestra
		110	Rock band
		120	Aircraft takeoff
		146	Threshold of pain

Effects of noise pollution on physical health

- The most direct harmful effect of excessive noise is **physical damage to the ear and the temporary or permanent hearing loss often called a temporary threshold shift (TTS)**. People suffering from this condition are unable to detect weak sounds. However hearing ability is usually recovered within a month of exposure.
- **Permanent loss, usually called noise induced permanent threshold shift (NIPTS)** represents a loss of hearing ability from which there is no recovery.

Effects of noise pollution on mental health:

- Noise can also cause **emotional or psychological effects** such as irritability, anxiety and stress.
- **Lack of concentration and mental fatigue** are significant health effects of noise.



Noise Control techniques

There are **four fundamental** ways in which noise can be controlled:

- **Reduce noise at the source,**
- **Block the path of noise,**
- **Increase the path length and protect the recipient.**
- **In general, the best control method is to reduce noise levels at the source.**

NUCLEAR/RADIATION POLLUTION

- Any undesirable effect caused to the environment due to radioactive substances or radiations is called **nuclear pollution**.
- Radioactive substances undergo natural radioactive decay in which unstable isotopes spontaneously give out fast moving particles, high energy radiations till a new stable isotope is formed. The isotope release energy either in the form of γ -rays or ionization process (α and β -particles).
- Radiations emitting from radioactive materials and the disposal of radioactive wastes cause **pollution** and damage the environment.

Sources:

Natural- cosmic rays from outer space; soil, rocks, air, water and food which contains one or more radioactive substances

Man-made/Anthropogenic – Nuclear power plants; Nuclear accidents; X-rays; diagnostic kits, laboratories where radioactive substances are used; Nuclear weapons

Effects of Nuclear pollution

- 1. Genetic mutations** - Radiation leads to **damage of DNA strands leading to genetic break up** in the course of time. The degree of genetic mutation leading to changes in DNA composition vary due to the level of radiation one has been exposed to and the kind of exposure. When a human or an animal is exposed to too much radiation from the atmosphere, once in the body, it remains active because energy cannot be destroyed. The **resulting mutation makes one highly susceptible to cancer**. For **pregnant women, kids born have adverse defects caused by genetic mutations** like low weight during birth. Effects such as disfigured births and impairment like blindness in children have also been reported. Infertility has also been mentioned as an effect of radiation.
- 2. Diseases** - Cancer is the most dominant radiation related disease. It has developed over the years and poses great risk in global health. Others include **leukemia, anemia, hemorrhage, a reduction in the life span leading to premature aging and premature deaths** as well as others such as cardiovascular complications



3. **Soil infertility** - Radioactive substances in the soil react together with the various nutrients leading to destruction of those nutrients, thus rendering the soil infertile and highly toxic. Such soil leads to the harvest of crops that are riddled with radiation and genetically modifies thus, unfit for consumption by both humans and animals. Since plants are at the base of the food chain, the herbivores consume them and retain the radiation levels which travels up in the food chain and food web, increasing their levels of radiation – explained through the concept of Biomagnification.
4. **Cell destruction** - Radioactive pollution has diverse effects such as the alteration of cells. Radiation distorts the cells present leading to **permanent damage of the various organs and organ systems**. In the face of too much radiation, permanent illnesses and death are inevitable.
5. **Burns** - Radiation is not easy to feel but it is easy to realize that you have been affected by it. The immediate **presence of burns, red lesions and sores is evidence**. This can **ultimately lead to skin cancer**.

Control of Nuclear Pollution

1. **Proper method of disposing radioactive waste** - Radioactive waste still has some level of radiation. So it cannot be disposed in the same way as normal waste. Since there is likelihood of seepage, this waste **should be stored in heavy and thick concrete containers**. Another option is to dilute the radiation since storage may not be possible.

2. **Proper labeling**

It is necessary for any material with **radioactive content to be labeled and the necessary precautions advised on the content of the label**. The reason for this is because radiation can enter the body by a mere touch of radioactive material. Containers with such elements should be well labeled in order for one to use protective gear when handling them.

3. **Banning of nuclear tests**

It has already been proven that nuclear power has a lot of latent power that is very destructive. Nevertheless, the tests done to perfect the energy contribute greatly to the overall presence of radioactive substances. Moreover, these tests though done in the deserts end up escaping from one ecosystem to another eventually affecting the lives of many people.

4. Alternative energy sources

The evolution and use of nuclear power was not a bad thing initially. However, considering the damage and threats it has on the environment, it is high time for its use to be discontinued and for the world to perhaps focus on alternative and **environmentally friendly energy sources** – like **renewable sources of energy** namely **Solar, hydro-electric and wind power**.

5. Proper storage

It is mandatory for containers carrying radioactive material to be stored properly. For starters, such substances should be stored in **radiation proof containers to ensure no seeping or leakage during handling**.

6. Reusing

Since it is not easy to store or dispose the waste, it can be recycled and used for other purposes like in another reactor as fuel thereby protecting the environment.



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AIR POLLUTION

- Air is one of the most important natural resources essential for the substance of living beings. It is a matter of serious concern that the ambient air is being contaminated day by day especially due to human activities.
- **Definition:** Prevention and control of pollution, Act 1981 defines air pollution as any solid, liquid or gaseous substance (including noise) present in the atmosphere in such concentration that may or tend to be injurious to human beings or other living beings or other living creature or plants or property or enjoyment.

SOURCES OF AIR POLLUTION

Natural sources:

- volcanic eruptions,
- forest fires,
- sea salt sprays,
- pollen grains and pores.
- Radioactive minerals present in the earth crust are the sources of radioactivity in the air.

Man-made/anthropogenic

sources : These pollutants are generated due to various industrial and domestic activities of man. These include

- thermal power plants,
- Industrial units,
- fossil fuel burning,
- vehicular activities and
- agricultural operations.
- Fertilizer units,
- Smelters,
- textile mills,
- refineries and chemical industries Automobile exhaust

TYPES OF AIR POLLUTANTS

According to

a) State of origin:

- **Primary pollutants:** The pollutants that are emitted directly from identifiable sources produced by natural events (eg: dust storms and volcanic eruptions) and human activities (eg: emissions from vehicles, industries etc.) are called primary pollutants. Eg: smoke, dust, oxides of sulphur & nitrogen, hydrocarbons and particulate matter etc.
- **Secondary pollutants:** The pollutants that are formed in the atmosphere by chemical interactions between primary pollutants and atmospheric constituents are known as secondary pollutants. Eg. Sulphur trioxide, ozone, ketones, sulphuric acid, nitric acid, carbonic acid etc.

b) According to state of matter

- **Gaseous air pollutants:** These pollutants exist in a gaseous state at normal temperature and pressure. They are carbon dioxide, nitrogen dioxide, sulphur oxides etc.
- **Particulate air pollutants:** These are not gaseous substances. They are suspended droplets, solid particles or mixtures of the two.

c) According to presence in environment

- **Indoor air pollutants:** paints, glues, varnishes etc.
- **Outdoor air pollutants:** mining, automobiles, industrial pollutants etc

Common air pollutants

1. Carbon monoxide (CO)

Source: incomplete combustion of fuels, automobile exhaust, industrial effluents, volcanic eruptions

Sink – a system which absorbs or store the pollutant, thereby nullifying its harmful effect.

Sink for CO: microorganisms present in the soil act as a major sink for CO; they convert CO to CO₂

Effects – affect the human health as it interferes combines with blood's ability to carry oxygen; its forms carboxy-haemoglobin which ultimately affects the brain function. Acute starvation of oxygen lead to coma and death

Control of CO

- **Modification of engine design-** engine should be modified so that right amount of O_2 is maintained for complete oxidation of fuel
- **Fuel-modification-** natural gas, blends of hydrocarbons must be used.
- **Treatment of exhaust gases-** two stage catalytic convertors are used to lower pollution



IInd stage – oxidising catalysts of noble metals (Pt, Ru) supported on ceramic materials are used to ensure oxidation of CO to CO_2

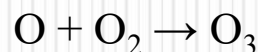
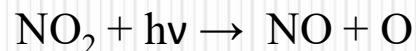
2. OXIDES OF NITROGEN (NO_x – NO and NO_2)

Sources –

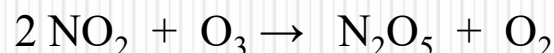
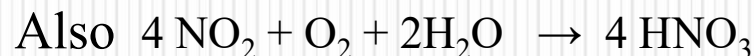
- **Natural** – during lightening discharge N_2 and O_2 combine to form NO; also formed by fixation of N from atmosphere or organic matter
- **Man-made** - Combustion of fossil fuels, from chemical industries as byproducts

Sink for NO_x

NO and NO_2 undergoes various photochemical and chemical reactions in the atmosphere leading to HNO_3 formation



Thus destruction of NO_2 and its regeneration is a cyclic process.



Effects of NO_x pollution

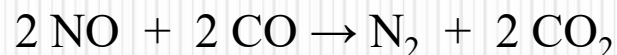
- No adverse health effect of NO are known at the conc. found in atmosphere.
- NO can oxidise to NO₂, which further reacts with hydrocarbons in the presence of sunlight to form photochemical smog
- NO₂ also forms nitric acid which lead to acid rain-decreases the pH of soil, corrodes marble, metals etc.
- Prolonged exposure leads to bronchitis

Control of NO_x

- **Modifying the engine design**
 - **Scrubbing the flue gases with sulfuric acid**
- $$\begin{aligned}\text{NO}_2 + \text{NO} &\rightarrow \text{N}_2\text{O}_3 \\ \text{N}_2\text{O}_3 + 2 \text{H}_2\text{SO}_4 &\rightarrow 2 \text{NOHSO}_4 + \text{H}_2\text{O} \\ 2 \text{NOHSO}_4 + 1/2 \text{O}_2 + \text{H}_2\text{O} &\rightarrow 2 \text{H}_2\text{SO}_4 + 2 \text{NO}_2 \\ 3 \text{NO}_2 + \text{H}_2\text{O} &\rightarrow 2 \text{HNO}_3 + \text{NO}\end{aligned}$$

NO and NO₂ again recirculated through scrubber.

- **Selective catalytic reduction:** NO_x is added to the exhaust gases and mixture is passed over a fixed bed catalyst such as CuO



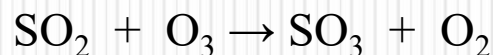
3. Oxides of sulfur SO_x – SO₂, SO₃

Sources-

- Volcanic eruptions, from petroleum refining, copper smelting ($\text{Cu}_2\text{S} + \text{O}_2 \rightarrow 2 \text{Cu} + \text{SO}_2$), cement manufacturing plants, burning fossil fuels ($\text{S} + \text{O}_2 \rightarrow \text{SO}_2$), photolytic oxidation of SO₂

Sinks for SO_x

- photolytic oxidation of SO₂ followed by catalytic reaction with ozone



- also leads to acid rain formation ; combines with limestone minerals which act as minerals for SO₂



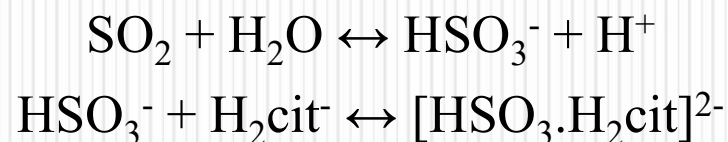
Effects

- cause of acid rain
- Causes chronic bronchitis
- Prolonged exposure causes damages to buildings, marble etc. ($\text{CaCO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + \text{CO}_2 + \text{H}_2\text{O}$)
- Chlorosis in plants, the formation of chlorophyll is lowed down, leaves turn yellow

Control of SO_x

- Use of fuels with less sulfur
- Use of chemical scrubbers for removing SO₂ from exhaust gases

Citric acid scrubber



the solution containing citrate ions is transferred to closed vessel and H₂S is passes, S precipitates out and citrate ions are regenerated

4. Particulates

- Any dispersed matter-solid or liquid in air, size ranging from a diameter of 0.002 to 500 μm ; May be organic (contains polyaromatic hydrocarbons) or inorganic (silicates, asbestos, lead halides, iron oxide etc.)
- Examples are Aerosols, dust, fumes, mist, flyash etc.

Sources – Burning of wood, fossil fuels, mining operations, industries, forest fires, dust storms, volcanoes etc.

Control of particulate emissions

- Gravity settling chamber
- Filters
- Scrubbers
- Cyclone collector

Harmful effects of particulates on

Human life

- Respiratory diseases
 - Silicosis
 - Asbestosis
 - lung cancer
 - Nervous disorders
 - Renal dysfunction
- Large particles enter the respiratory system and are trapped by lining of hair of the nose, but smaller particles move to the throat and captured by mucus membrane, but ultrasmall ones traverse upto the lungs and deposit there act as adsorption sites for carcinogens

Materials

- Accelerate corrosion of materials
- Fumes, dust, soot, aerosols cause damage to soil, buildings, monuments, sculptures.
- Smoke particles stick on every surface forming a film of soot

Climate

- Particulate help in maintaining the radiation balance and heat balance on earth
- Absorbs free radicals, act as free radical inhibitors
- Provide nuclei for condensation of water vapour leading to formation of cloud and smoke

5. HYDROCARBONS

Sources-

- Trees -
- Organic matter $\xrightarrow{\text{Bacteria}}$ $\text{CH}_4 + \text{CO}_2$
Soil, water
- Motor fuels
- volatile organic solvents used in paints

Sinks

Hydrocarbons undergo photochemical and chemical reactions to form water and carbon dioxides



Effects

- lung cancer
- Cause photochemical smog
- Inhalation of benzene, toluene causes irritation in mucus membrane
- Methane causes narcotic effect on humans

Control of hydrocarbons

- Being thermodynamically unstable, get oxidized in the atmosphere by a series of chemical and photochemical reactions which produces various byproducts like carbon dioxide, solid organic particulates, which settle down and acid are washed down by rain
- Hydrocarbons from automobile exhaust are controlled by Incineration (completes the oxidation) or Adsorption (on a bed of carbon or through a liquid in which hydrocarbons dissolve)

Environmental effects of air pollution

All the pollutants discussed earlier are posing a great threat to the Earth's environment

Various environmental problems are

- **Photochemical smog**
- **Acid rain**
- **Global warming**
- **Ozone depletion**

SMOG: fog condensing on the carbon particles of smoke

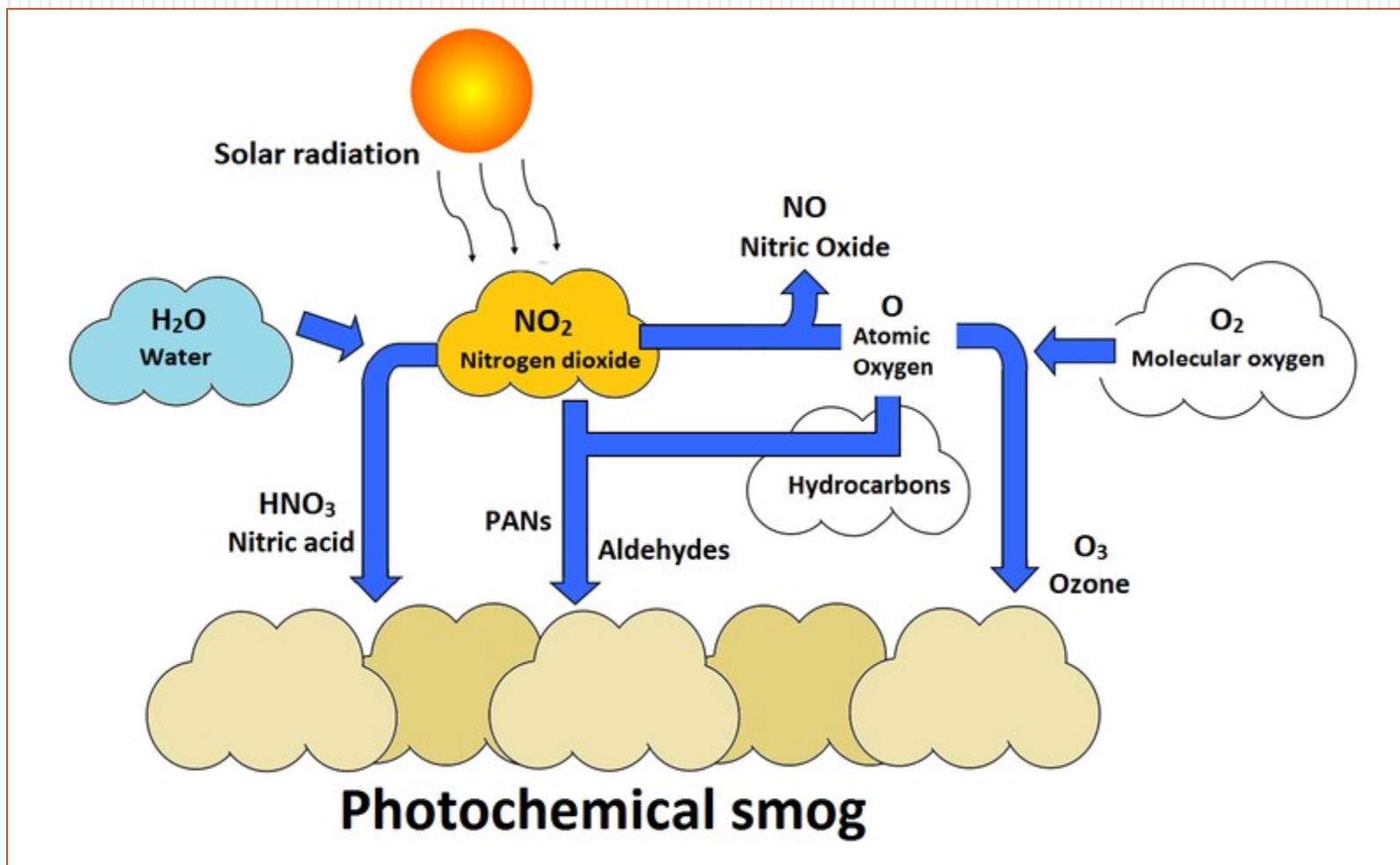
Smog is of two types depending upon the composition, or method of its formation or the place famous for its formation:

Classical smog	Photochemical smog
1. This type of smog is formed due to high concentration of SO_2 and particulate matter produced from fuel combination.	1. This type of smog is formed due to the action of sunlight on nitrogen oxides and hydrocarbons produced by factories or automobiles.
2. It is reducing in nature.	2. It is <u>oxidising</u> in nature.
3. It occurs in cool, humid climate.	3. It occurs in warm, dry and sunny climate.
4. It can cause irritation, bronchitis and lung problem.	4. It causes irritation to the eyes.

5. Also known as LONDON smog

5. Called LOSANGELES smog

Chemistry of Photochemical Smog - involves free-radical mechanism



Note - Detailed mechanism to be studied from textbook

ACID RAIN

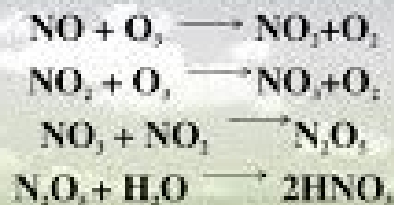
- Acid rain results when sulfur dioxide (SO_2) and nitrogen oxides (NO_x) are emitted into the atmosphere and transported by wind and air currents. The SO_2 and NO_x react with water, oxygen and other chemicals to form sulfuric and nitric acids. These then mix with water and other materials before falling to the ground.

Chemical Processes Involved In acid rain

Formation Of Sulphuric Acid



Reaction Involving Formation Of Nitric Acid



Effects of Acid rain

- Acid rain runoff from catchment areas into rivers and lakes has also reduced biodiversity as rivers and lakes become more acidic.
- Damage to the vegetation
- Damage to building and sculptures
- Acid rain impacts soil chemistry and biology adversely
- Acid rain leads to weathering of buildings, corrosion of metals, and peeling of paints on surfaces.
- Human health is not directly affected by acid rain because acid rain water is too dilute to cause serious health problems. However, the dry depositions also known as gaseous particulates in the air which in this case are nitrogen oxides and sulfur dioxide can cause serious health problems when inhaled. Intensified levels of acid depositions in dry form in the air can cause lung and heart problems such as bronchitis and asthma.



Global Warming

- Global warming, a phenomenon which is caused due to the increase in concentration of certain gases like carbon dioxide, nitrogen oxides, methane and CFCs. Observations of the earth have shown that atmospheric constituents such as water vapour, carbon dioxide, methane, nitrogen oxides and Chloro Fluoro Carbons trap heat in the form of infra-red radiation near the earth's surface. This is known as the 'Greenhouse Effect'.

Impacts of global warming

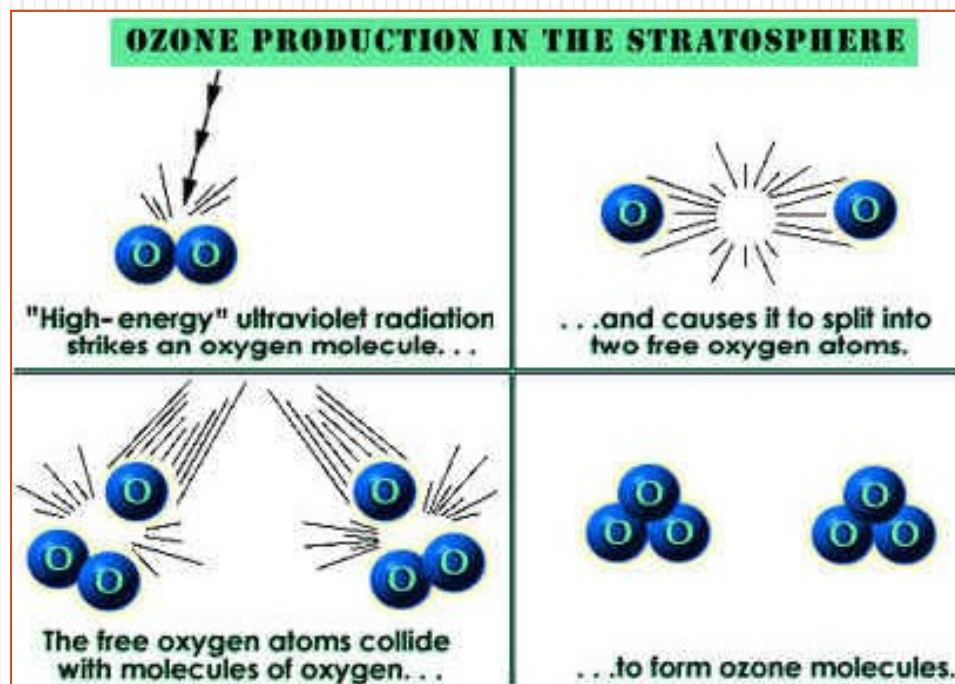
- Melting of glaciers and polar ice caps, rising sea levels thereby submerging the coastal areas.
- Adverse affect on food production in the world
- Drastic change in the climate-which will alter crop yields, water supply
- May initiate forest fires

Remedial measures

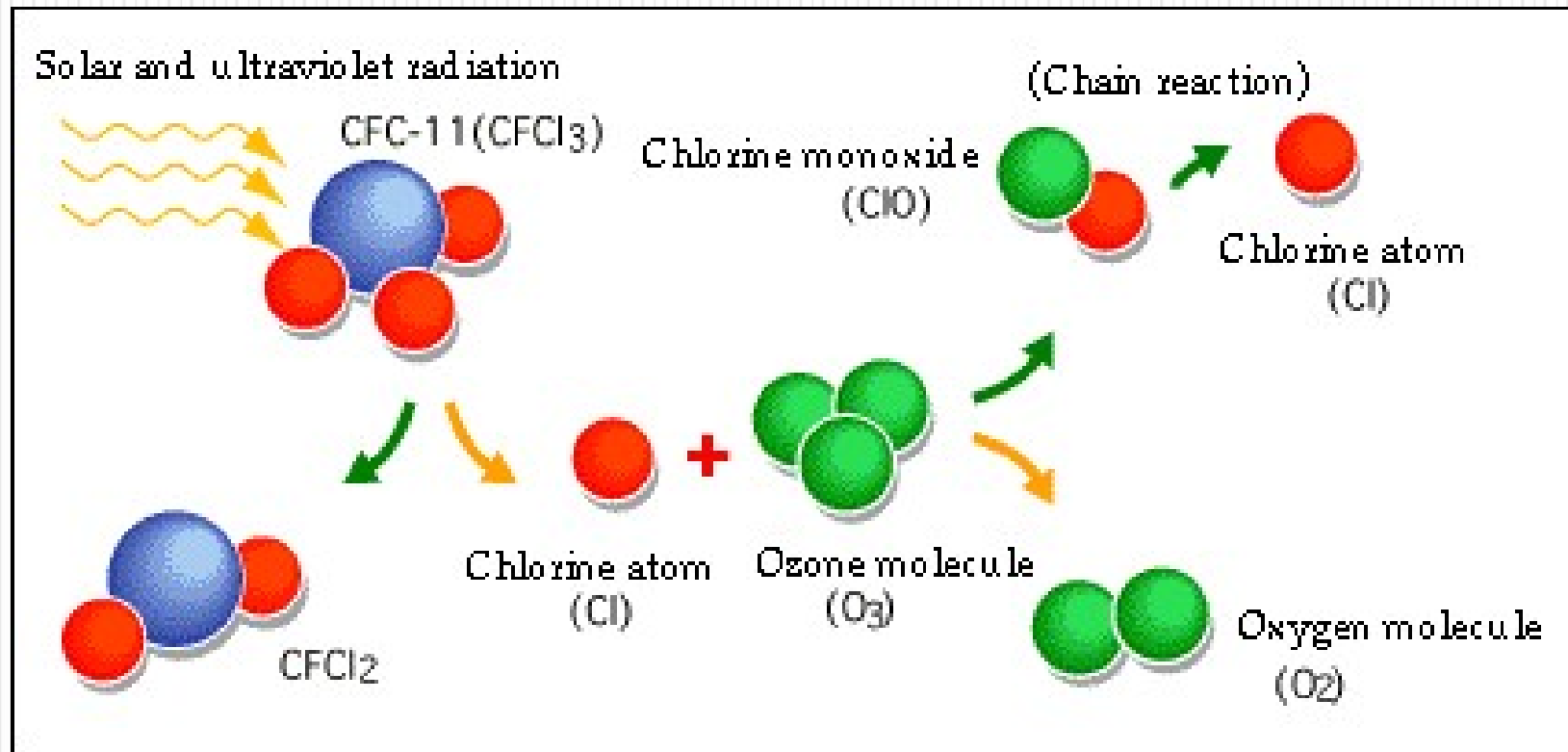
- Reducing consumption of fossil fuels
- Reduction in CFC production
- Reforestation and conservation of green areas
- Recovering Green house gases from the atmosphere

Ozone layer depletion

- Ozone depletion**, gradual thinning of Earth's ozone layer in the upper atmosphere caused by the release of chemical compounds containing gaseous chlorine or bromine from industry and other human activities. The thinning is most pronounced in the polar regions, especially over Antarctica. Ozone depletion is a major environmental problem because it increases the amount of ultraviolet (UV) radiation that reaches Earth's surface, which increases the rate of skin cancer, eye cataracts, and genetic and immune system damage.



OZONE LAYER DEPLETION MECHANISM



Hydrofluorocarbons (HFCs) and Hydrochlorofluorocarbons (HCFCs) are substitutes for CFCs. They are chemically removed from the troposphere and hence don't cause ozone depletion.



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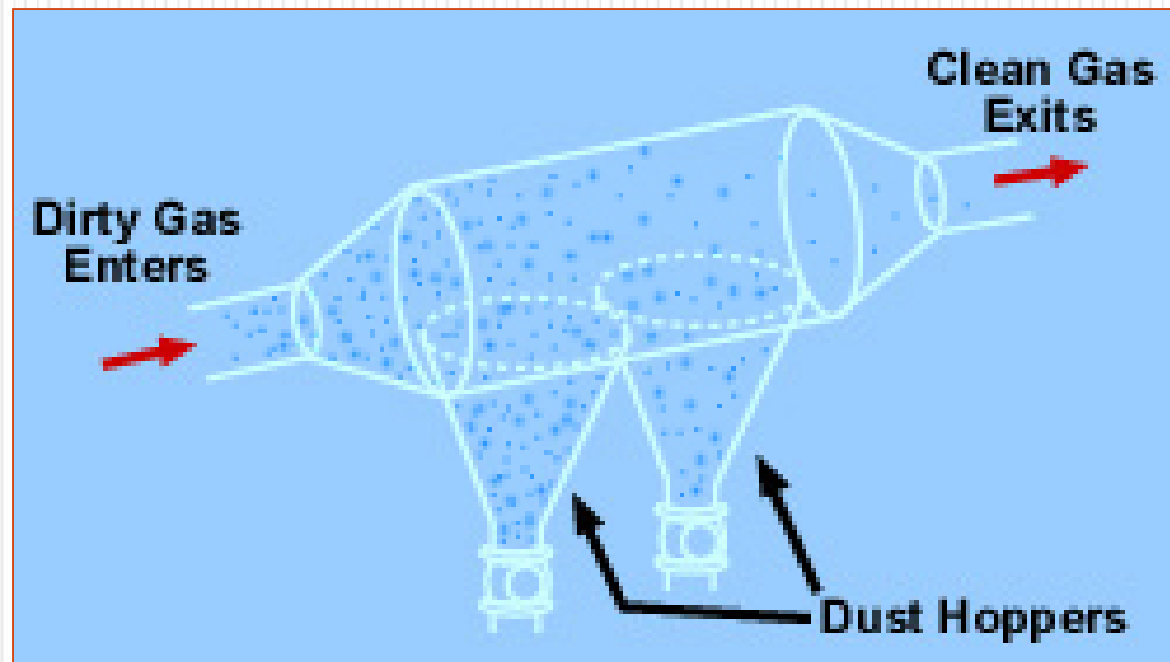


AIR POLLUTION CONTROL METHODS

Gravity settling chambers / Gravitational settling

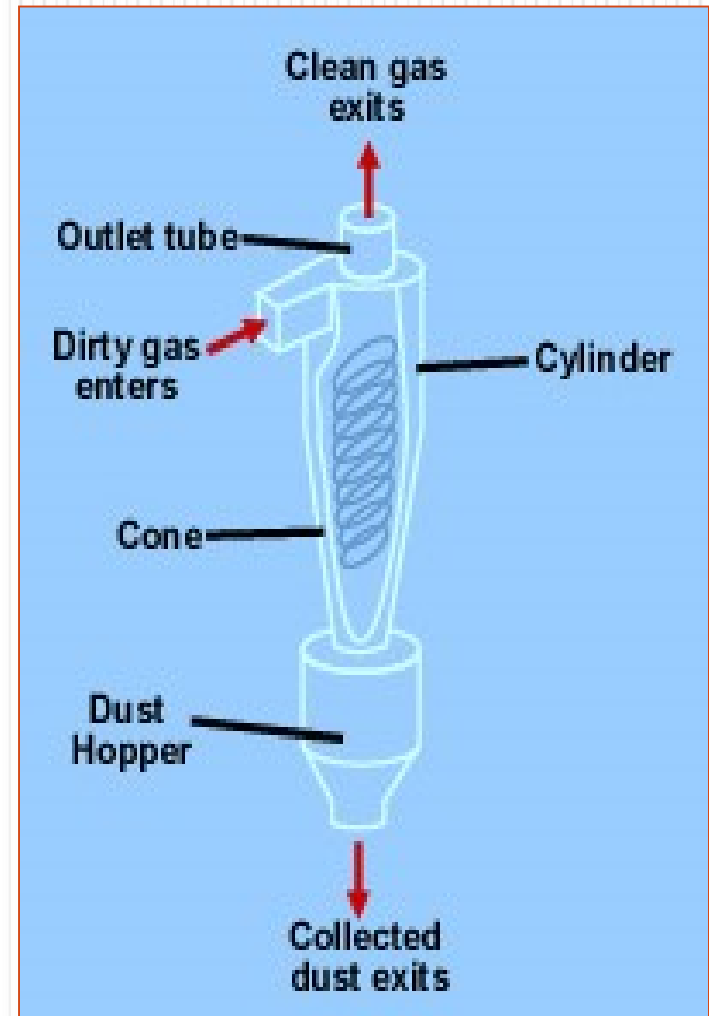
Settling chambers use the **force of gravity** to remove solid particles $>50\text{ }\mu\text{m}$.

The gas stream enters a chamber where the **velocity of the gas is reduced**. Large particles drop out of the gas and are recaptured in hoppers. Because settling chambers are effective in **removing only larger particles**, they are used in conjunction with a more efficient control device.



Cyclone collectors

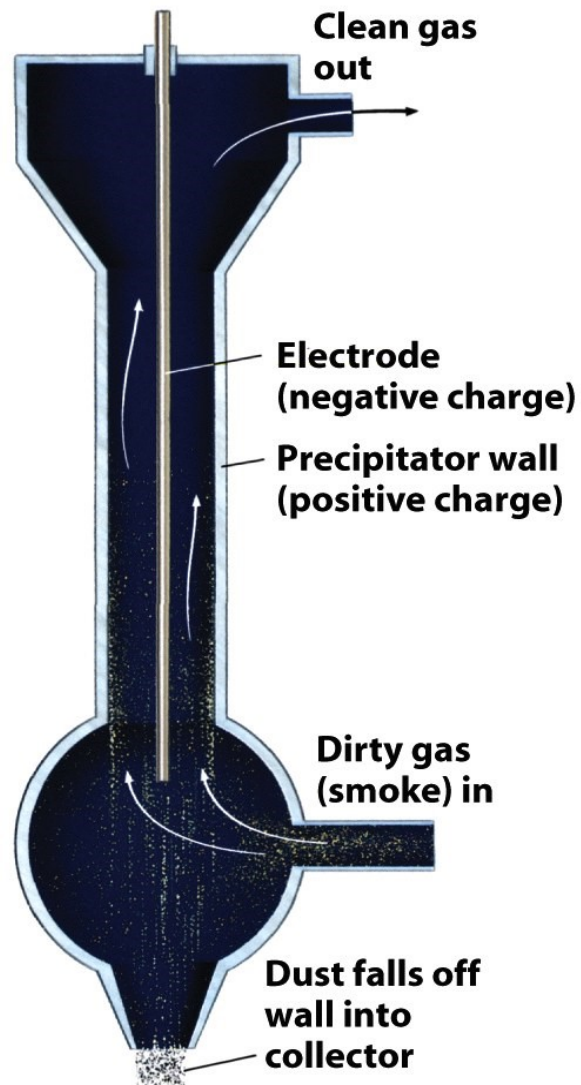
- The general **principle of inertia separation** is that the particulate-laden gas is forced to change direction. As gas changes direction, the **inertia of the particles causes them to continue in the original direction** and be separated from the gas stream.
- The **walls of the cyclone narrow toward the bottom** of the unit, allowing the particles to be collected in a hopper.
- The **cleaner air leaves** the cyclone through the top of the chamber, **flowing upward in a spiral vortex**, formed within a downward moving spiral.
- Cyclones are **efficient in removing large particles** but are not as efficient with smaller particles. For this reason, they are used with other particulate control devices.



Electrostatic precipitator

- It is a type of filter (dry scrubber) that uses static electricity to remove soot and ash from exhaust fumes before they exit the smokestacks. Smoke consists of tiny particles of soot that are suspended in hot, rising air. These unburned particles of carbon are pulled out of the smoke by using static electricity in the precipitators, leaving clean, hot air to escape the smokestacks.
- **Working:** The dirty flue gas escaping through the smokestack is passed through two electrodes. These electrodes can be **metal wires, bars, or plates** inside a pipe or the smokestack itself. One of the electrodes is charged with a high negative voltage, and this plate causes particulates inside the smoke to obtain a negative charge as they pass by this electrode. Further along the pipe, the second electrode carries a similarly high positive voltage. The negatively charged soot particles are pulled towards the positive electrode and stick to it. Occasionally these plates must be cleaned to remove the accumulated soot and dispose of it into a hopper.

Electrostatic precipitator

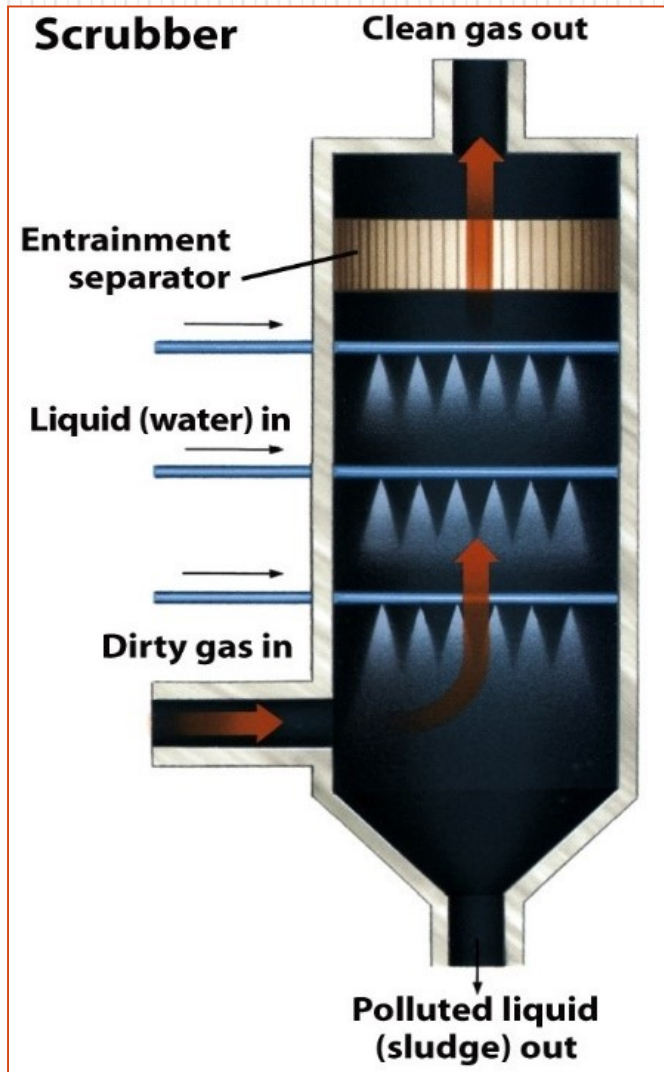


WET SCRUBBERS

- **Wet Scrubbers** are effective **air pollution control** devices for removing particles and/or gases from industrial exhaust streams.
- operates by introducing the dirty gas stream with a **scrubbing liquid** – typically water.

Other Ways to Improve Air Quality

- **Reduce sulfur content in gasoline** from its current average of 330 ppm to 30 ppm as sulfur clogs catalytic converters
- Require **federal emission standards** for all passenger vehicles Including SUVs, trucks and minivans
- Require **emission testing** for all vehicles Including diesel.





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WATER POLLUTION

- Most precious gift of nature, vital for the survival of all living beings
- 71% of the earth's surface is covered by water only a tiny fraction of this water is available to us as fresh water. About 97% of the total water available on earth is found in oceans and is too salty for drinking or irrigation. The remaining 3% is fresh water. Of this 2.997% is locked in ice caps or glaciers. Thus only 0.003% of the earth's total volume of water is easily available to us as soil moisture, groundwater, water vapour and water in lakes, streams, rivers and wetlands.
- **When the quality or composition of water changes directly or indirectly as a result of man's activities such that it becomes unfit for any purpose it is said to be polluted.**



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Causes of Water pollution

- **Point sources** of pollution: When a source of pollution can be readily identified because it has a definite source and place where it enters the water, it is said to come from a point source. Eg. Municipal and Industrial Discharge Pipes.
- **Non-point sources** - When a source of pollution cannot be readily identified, such as agricultural runoff, acid rain, etc,



- Disease-causing agents, oxygen depleting wastes, inorganic plant nutrient, water soluble inorganic chemicals, organic chemicals, Sediment of suspended matter, Oil, Water soluble radioactive isotopes, Hot water let out by power plants,
- Groundwater pollution- Urban run-off of untreated or poorly treated waste water and garbage
- Industrial waste storage located above or near aquifers
- Agricultural practices such as the application of large amounts of fertilizers and pesticides, animal feeding operations, etc. in the rural sector.
- Leakage from underground storage tanks containing gasoline and other hazardous substances
- Leachate from landfills
- Poorly designed and inadequately maintained septic tanks
- Mining wastes

Control of Water Pollution

Pollution of water bodies can be controlled through practicing following measures –

- I. The catchment and feeders of the rivers, lakes ponds and streams should be **regularly monitored**
- II. Industrial plants should follow **recycling operations**.
- III. Sewage and industrial **effluents** should be **properly treated through physical, chemical and biological processes** to minimize pollution load prior to release it in water body.
- V. Wastes containing high total dissolve solids (TDS) should be treated for **separation of liquid and residue**.



VI. Treated waste water can be used as a cooling water make up. **Cooling water can be processed or stored in artificial ponds** until the difference in temperature between it and receiving water is nearly equal.

VII. Water containing radioactive substances should be treated separately by means of **dewatering procedure**.

VIII. The most important and effective solution for environmental pollution control lies in **social awareness and public participation**.



Effects of water pollution

- **Water born diseases** especially in developing countries like cholera, tuberculosis
- The **inorganic and organic pollutants in water** are responsible for **reduction of species diversity** in the system.
- Suspended solids such as silt and coal may injure the gills of the fish and cause asphyxiation.
- The **oxygen depletion** due to pollution load and **eutrophication** leads to death of aquatic animals.
- **Water bodies** receiving industrial and municipal effluent – indicate various physical changes too. e. g., **color of water and offensive odour**. This prevents growth of fresh water plants and animals.
- Polluted water having **acidic or alkaline nature corrodes concrete and metal pipes**.



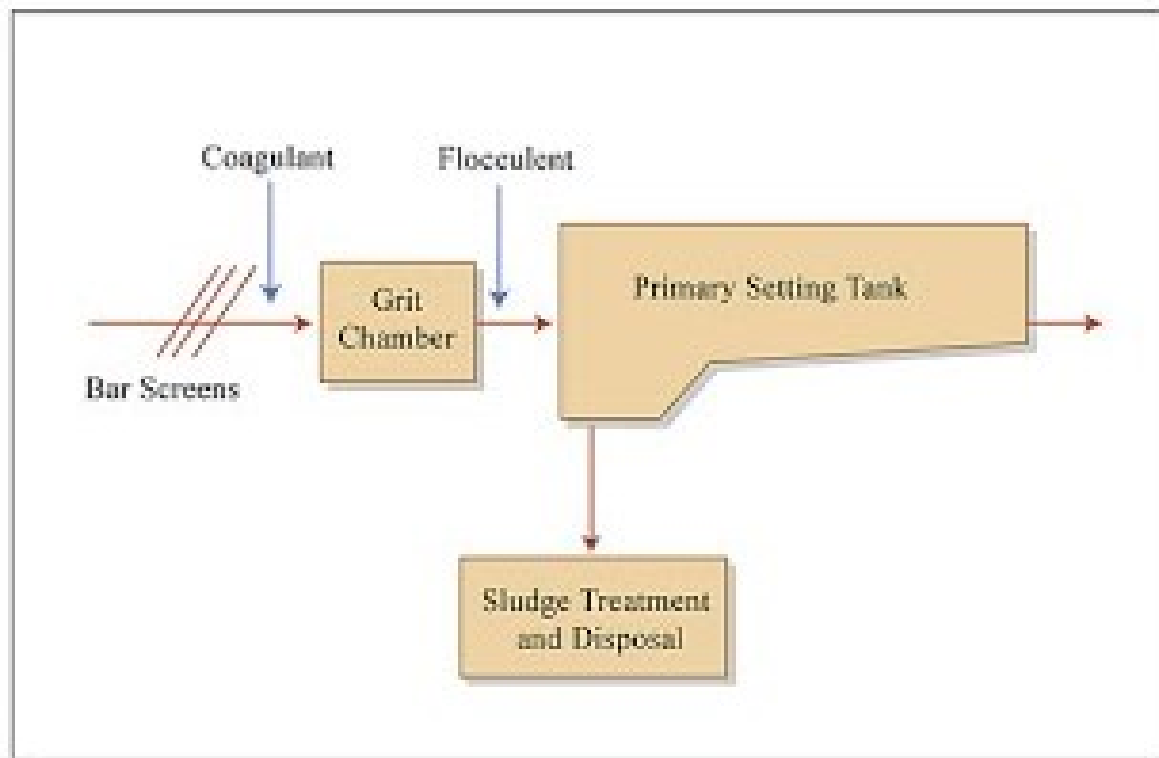
CHARACTERISTICS OF WASTE WATER

1. **Biochemical oxygen demand (BOD)** represents the amount of oxygen consumed by bacteria and other microorganisms while they decompose organic matter under aerobic (oxygen is present) conditions at a specified temperature.
2. **Chemical oxygen demand (COD)** refers the requirement of dissolved oxygen for both the oxidation of organic and inorganic constituents by strong oxidizing reagents. **COD** levels in waste streams are always higher than **BOD**.
3. **Dissolved oxygen (DO)** - is a measure of how much oxygen is dissolved in the water - the amount of oxygen available to living aquatic organisms.
4. **TDS**
5. **Free chlorine**
6. **Dissolved chlorides**
7. **Dissolved carbon dioxide**

WASTE WATER TREATMENT PROCESSES

- The waste water treatment processes are grouped according to the water quality they are expected to produce
- 1. **Primary treatment** – removes suspended solids and organic matter
- 2. **Secondary treatment** – involves biological process and organic matter is removed.
- 3. **Tertiary treatment** - Physical, chemical and biological processes depending upon the impurities

- **Primary Treatment** is the first process that takes place, it's usually the operation performed to remove those materials that separate easily. The first stage of waste water treatment, consisting of the **removal of a substantial amount of suspended matter by using, Sedimentation, Flocculation, Screening or similar methods.**



- a) **Screening**, removes **relatively large solids** in a primary treatment facility. During the Screening Treatment, **wood, dead animals, and rocks** get in the way and have to be removed to keep the process going.

- b) **Sedimentation** takes place in a **settling tank or a clarifier**. It allows many suspended particles to settle out of water or waste water while slowly going through the tank and providing a source of purification. A large amount of sludge is usually found at the bottom of the tank and is periodically removed.

- c) **Flocculation** is the action of **binding large particles into clumps**. It's where Colloids form into flakes or floc, colloids are usually dissolved into a liquid, they do this by gently mixing the coagulated water this is also called the hydrodynamic process. Flocs or flakes are supposed to grow to either 0.1 or 2.0mm so that it can be removed by sedimentation.

SECONDARY TREATMENT OF WATER

A treatment process for wastewater (or sewage) to achieve a certain degree of effluent quality by using a sewage treatment plant with physical phase separation to remove settleable solids and a biological process to remove dissolved and suspended organic compounds.

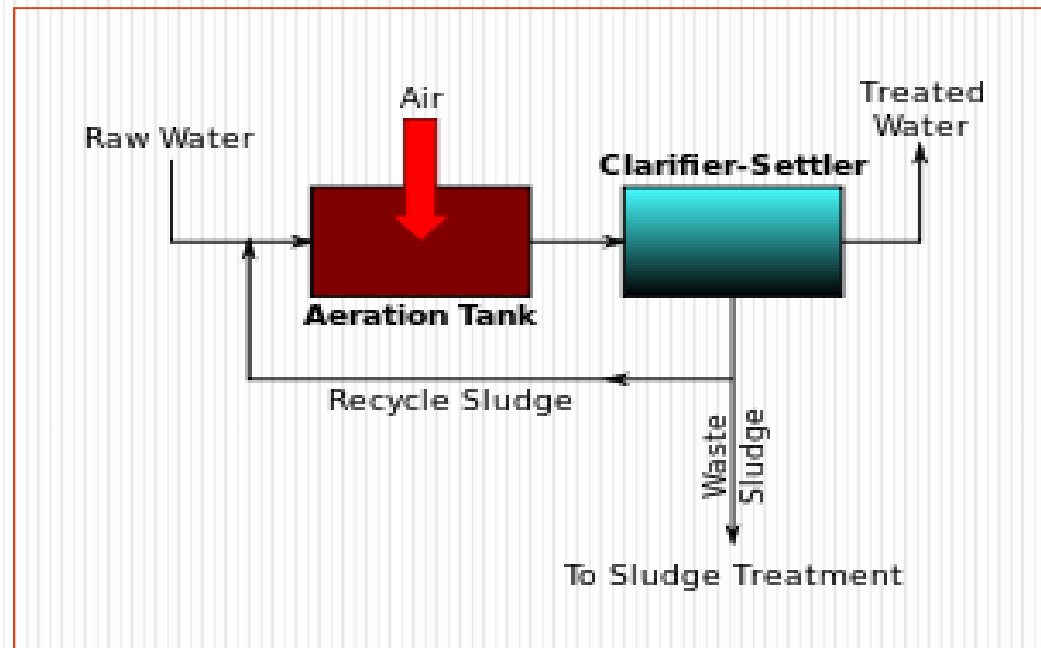
1. **Activated Sludge process**- The activated sludge process is a type of wastewater treatment process for **treating sewage or industrial wastewaters using aeration and a biological floc composed of bacteria and protozoa**. During the activated sludge process, primary effluent flows into an aeration tank, where it is mixed with microorganisms. The aeration tank injects a steady supply of oxygen or air into the wastewater, ensuring that the organisms have an adequate supply of oxygen needed to breakdown the organic matter that remains in the effluent.

Pros:

- Low construction cost
- Occupies small area
- Relatively low odour
- Removes a high percent of BOD

Cons:

- High operating cost (air pumps)
- High energy expenses for oxygen demand



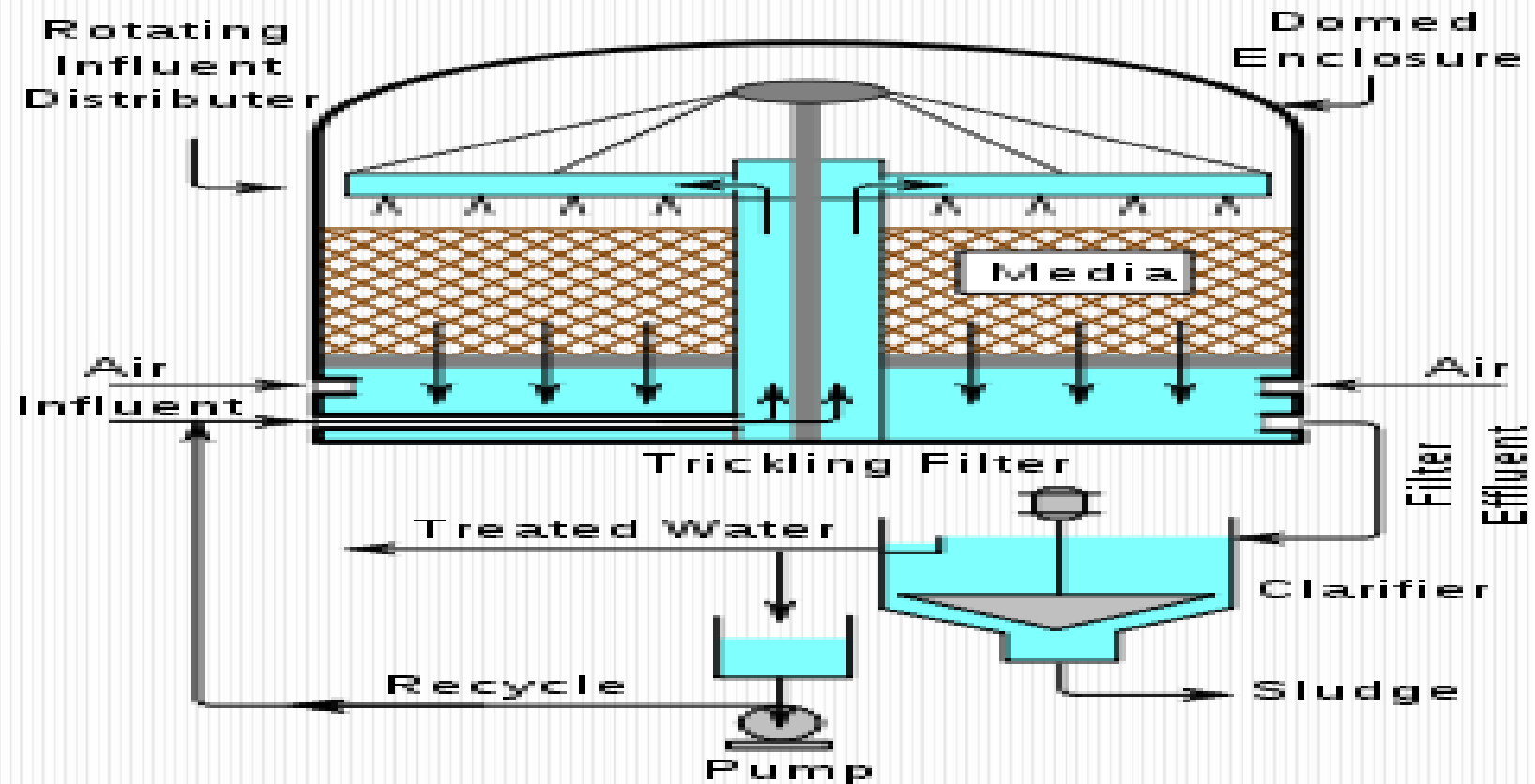
2. **Trickling filters** - A trickling filter is a type of wastewater treatment system. It consists of a **fixed bed of rocks, coke, gravel, slag, polyurethane foam, sphagnum peat moss, ceramic, or plastic media** over which sewage or other wastewater flows downward and **causes a layer of microbial slime to grow**, covering the bed of media.

Cons:

- Temperature Dependent
- Vulnerable to congestion, which can lead to flooding and system failure, low oxygen supply, and restricted water flow
- Can occupy a larger area than activated sludge

Pros:

- Low construction cost
- Cheap oxygen delivery
- Non-electric systems available



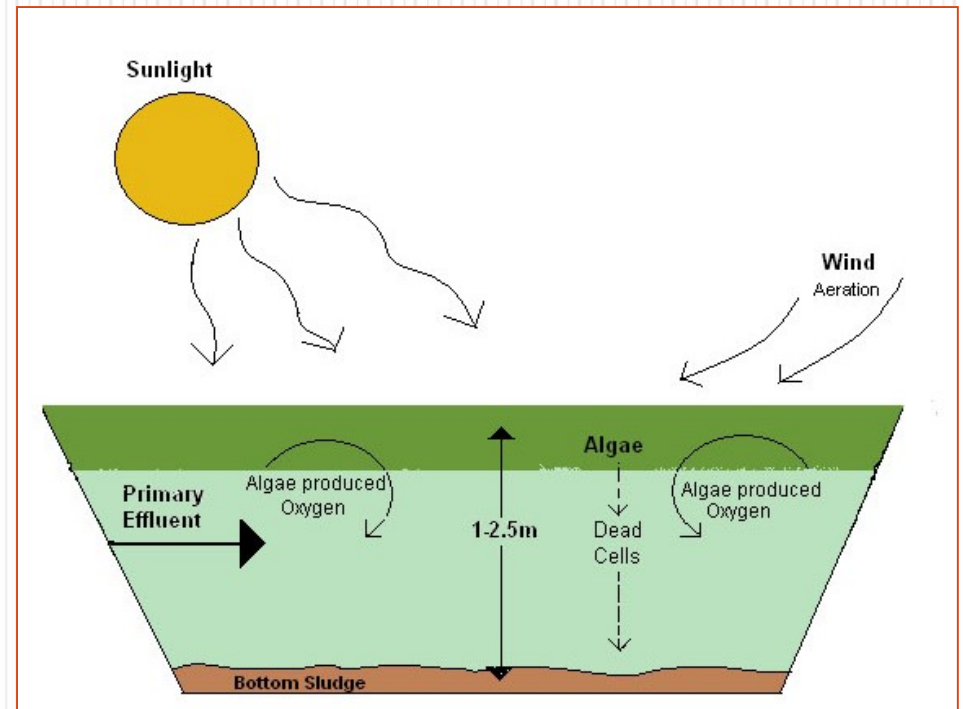
- **OXIDATION PONDS-** Oxidation ponds are large and shallow; a typical depth would range from 1-2.5m. The ponds are composed of microorganisms, which feed on the organic matter received from primary effluent. Algae are a key feature in the oxidation pond system; they deliver a steady flow of oxygen to the bacteria. The algae require sunlight to produce oxygen via photosynthesis, reaeration created by wind delivers air flow when sunlight is not available. Overall the process is slow and requires large areas of land.

Pros:

- Small energy input
- Degrades nitrogen and phosphorus

Cons:

- Occupies a large area
- Possible odors
- Slow process
- Long retention times
- Climate dependent



TERTIARY TREATMENT OF WATER

- **Tertiary treatment** is the **final cleaning process** that improves wastewater quality before it is reused, recycled or discharged to the environment. The treatment removes remaining inorganic substances, such as the nitrogen and phosphorus.
- First stage of tertiary treatment is **FILTRATION/MICROTRAINING** which helps primarily to **remove residual suspended matter** in wastewater. **Sand filtration or finely woven steel fabric** is the usually used for this purpose.

ADSORPTION- In some cases, residual toxins may be present in wastewater and to filter them out, **activated carbon is used to adsorb the toxins and remove them.**

- After filtration, **some fine particulate matter may still remain** in the wastewater. To treat the wastewater further, it is **transferred to lagoons with filter feeders such as *Daphne***. In the lagoon's aerobic and biologically enriched environment, further removal of fine particulate matter takes place.

- The wastewater may still have high nutrient concentration. **Nitrogen and phosphorus are usually found in high concentration in wastewater** and this, if untreated and released into the natural water environment, can cause eutrophication.
- **Both nitrogen and phosphorus can be removed using biological processes involving different bacteria.** Nitrogen in the form of ammonia is first oxidized to form nitrates and then nitrates to nitrogen gas which is released to the environment. On the other hand, **phosphorus** can be removed biologically as well by **chemical precipitation with salts of iron, aluminium, or lime.**
- The **FINAL STEPS in tertiary wastewater treatment**, before release into the environment, are **removing any undesirable microbes through the process of disinfection and also removal of odors from wastewater.** The clear and less cloudy nature of wastewater at this stage is critical for efficiency and effectiveness of the disinfection process. **Several disinfection agents can be used depending on wastewater condition (pH, clarity etc.) And among them chlorine, ozone, ultraviolet (UV) light are most common.**

SOLID WASTE AND ITS MANAGEMENT

- It is the **collecting, treating, and disposing of solid material** that is discarded because it has served its purpose or is no longer useful.
- **Improper disposal** of municipal solid waste can create **unsanitary conditions**, which can lead to environment pollution and to **outbreaks of vector-borne diseases**.
- Higher standards of living of ever increasing population has resulted in an increase in the quantity and variety of waste generated.

Therefore, management of solid wastes has become very important in order to minimize their adverse affects on the environment.

Sources/Classification of solid wastes

- (i) **Domestic/Residential Waste**: waste generated as a consequence of household activities such as cooking, cleaning, repairs, hobbies, redecoration, empty containers, packaging, clothing, old books, writing/new paper, and old furnishings.
- (ii) **Municipal Waste**: Wastes resulting from municipal activities and services such as street waste, dead animals, market waste and abandoned vehicles.
- (iii) **Commercial Waste**: solid wastes that originate in offices, wholesale and retail stores, restaurants, hotels, markets, warehouses and other commercial establishments.
- (i) **Institutional Waste**: Institutional wastes are those arising from institutions such as schools, universities, hospitals and research institutes.



- (v) **Garbage**: Includes **animal and vegetable wastes resulting from the handling, storage, sale, preparation, cooking and serving of food**. Such wastes contain putrescible organic matter, which produces strong odours and therefore attracts rats, flies. It requires immediate attention in its storage, handling and disposal.
- (vi) **Rubbish**: It is a general term applied to **solid wastes originating in households, commercial establishments and institutions, excluding garbage and ashes**.
- (vii) **Ashes**: Ashes are the residues from the burning of wood, coal, charcoal, coke and other combustible materials, for cooking and heating in houses, institutions and small industrial establishments. **When produced in large quantities at power generating plants and factories these wastes are classified as industrial wastes**. Ashes consist of a fine powdery residue, cinders and clinker often mixed with small pieces of metal and glass.
- (viii) **Bulky Wastes**: Wastes which **cannot be accommodated in the normal storage containers of households**. For this reason they require special collection. Examples of bulky wastes are large household appliances such as **cookers, refrigerators and washing machines as well as furniture, crates, vehicle parts, tyres, wood, trees and branches**.



- (ix) **Street Sweeping**: wastes that are collected from streets, walkways, alleys, parks and vacant lots. It commonly takes place in developing countries, where littering of public places is a far more widespread and acute problem. **Mechanised street sweeping** is the dominant practice in the developed countries. Street wastes include **paper, cardboard, plastic, dirt, dust, leaves** and other vegetable matter.
- (x) **Dead Animals**: This is a term **applied to dead animals that die naturally or accidentally killed**. This category does not include carcass and animal parts from slaughterhouses which are regarded as industrial wastes.
- (xi) **Construction and Demolition Wastes**: waste materials generated by the construction, refurbishment, repair and demolition of houses, commercial buildings and other structures. It mainly consists of **earth, stones, concrete, bricks, lumber, roofing materials, plumbing materials, heating systems and electrical wires** and parts of the general municipal waste stream, but when generated in large amounts at building and demolition sites, it is generally removed by contractors for filling low lying areas and by urban local bodies for disposal at landfills.



- (xii) **Industrial Wastes**: Discarded solid materials of manufacturing processes and industrial operations. They cover a vast range of substances which are unique to each industry. For this reason they are considered separately from municipal wastes.
- (xiii) **Hazardous Wastes**: Hazardous wastes may be defined as wastes of industrial, institutional or consumer origin which, because of their physical, chemical or biological characteristics are potentially dangerous to human and the environment. In some cases although the active agents may be liquid or gaseous, they are classified as solid wastes because they are confined in solid containers. Typical examples are: **solvents, paints and pesticides** whose spent containers are frequently mixed with municipal wastes and become part of the urban waste stream. Others, such as **pathological wastes from hospitals and radioactive wastes**, require special handling at all time.
- (xiv) **Sewage Wastes**: The **solid by-products of sewage treatment** are classified as sewage wastes. They are derived from the treatment of organic sludge from both the raw and treated sewage. The inorganic fraction of raw sewage such as grit is separated at the preliminary stage of treatment, but because it entrains putrescible organic matter which may contain pathogens, must be buried/disposed off without delay.



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Effects of solid waste

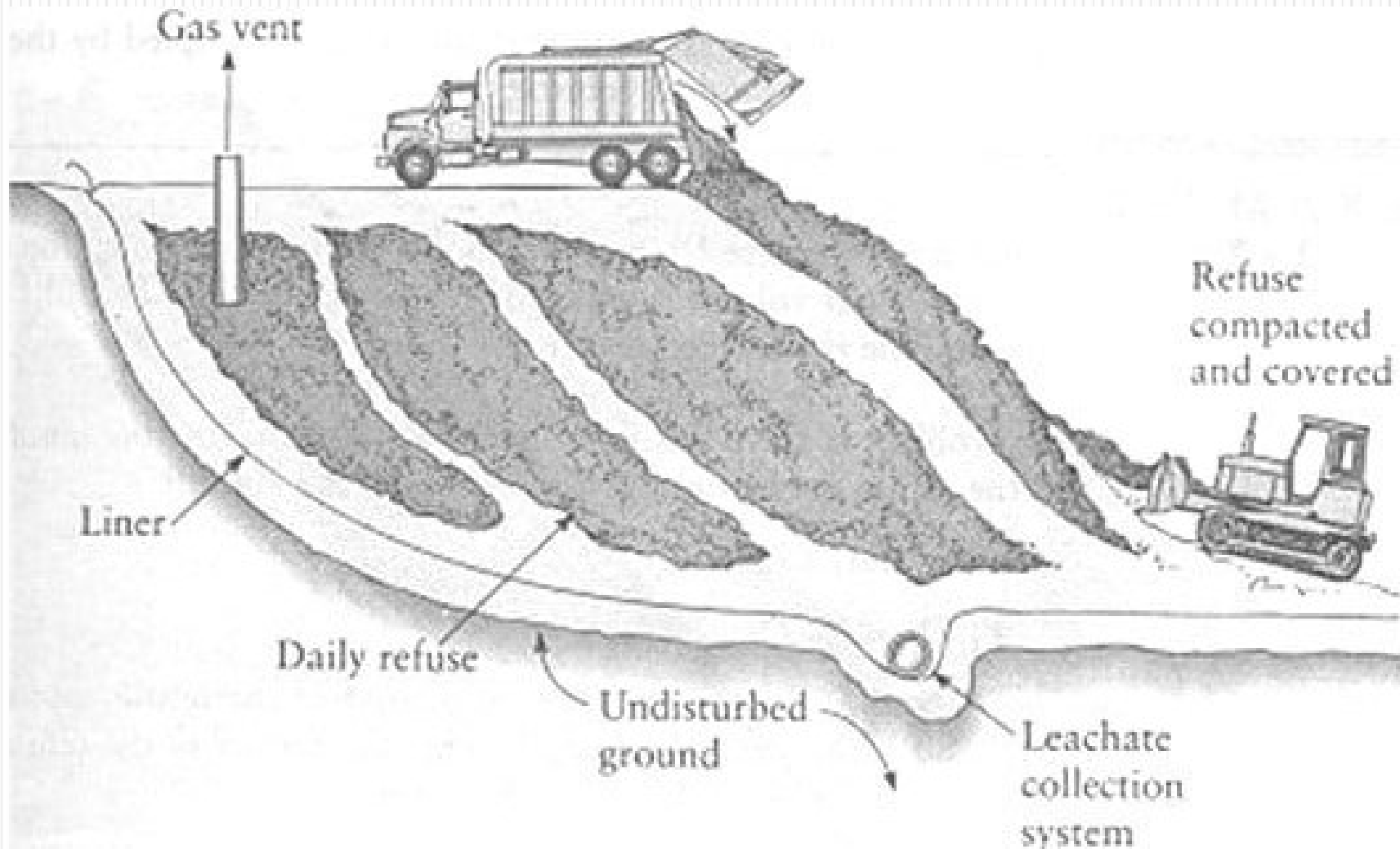
- Municipal solid wastes **heap up on the roads** due to improper disposal system.
- Dumping allows biodegradable materials to decompose under uncontrolled and unhygienic conditions which **produces foul smell and breeds various types of insects and infectious organisms** besides spoiling the aesthetics of the site.
- Industrial solid wastes contains **toxic materials**, which may spread on land and can cause **changes in physicochemical and biological characteristics** thereby affecting productivity of soils.
- Toxic substances may **leach or percolate** to **contaminate the ground water**.
- Various types of wastes like **cans, pesticides, cleaning solvents, batteries (zinc, lead or mercury), radioactive materials, plastics and e-waste** are **mixed up with paper, scraps and other non-toxic materials** which could be recycled. Burning of some of these materials produces **dioxins, furans and polychlorinated biphenyls**, which have the potential to cause various types of ailments including cancer.

Treatment and disposal methods

- The selection of the disposal method upon the type and source of solid waste, as the composition of solid waste is not same everywhere
- Most prominent solid waste disposal methods are :
 1. **Sanitary landfill**
 2. **Incineration**
 3. **Composting**
 4. **Dumping**
 5. **Recycling and reuse**

Sanitary landfills

- In a sanitary landfill, garbage is spread out in thin layers, compacted and covered with clay or plastic foam. In the modern landfills the bottom is covered with an impermeable liner, usually several layers of clay, thick plastic and sand. The liner (impermeable material like plastic) protects the ground water from being contaminated due to percolation of leachate. Leachate from bottom is pumped and sent for treatment. **When landfill is full it is covered with clay, sand, gravel and top soil to prevent seepage of water.**
- Several wells are drilled near the landfill site to monitor if any leakage is contaminating ground water.
- **Methane** produced by anaerobic decomposition is collected and **burnt to produce electricity or heat.**

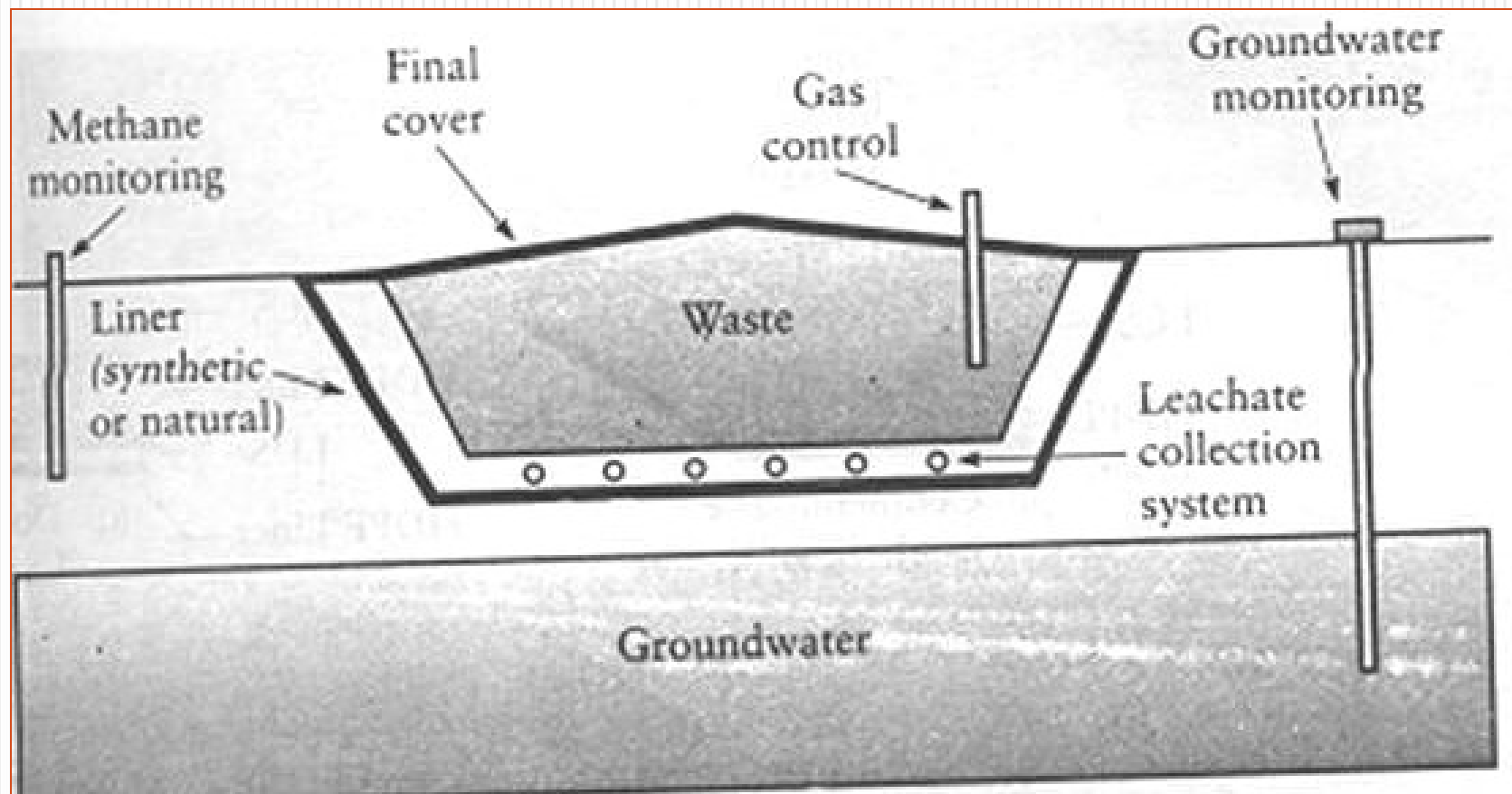


Adverse impacts may occur from landfill operations.

These impacts can vary:

- i. **Fatal accidents** (e.g., scavengers buried under waste piles).
- ii. **Infrastructure damage** (e.g., damage to access roads by heavy vehicles).
- iii. **Pollution** of the local environment (such as contamination of groundwater and/or aquifers by leakage and residual soil contamination during landfill usage, as well as after landfill closure).
- iv. Off gassing of **methane** generated by decaying organic wastes (methane is a **greenhouse gas** many times more potent than carbon dioxide, and can itself be a danger to inhabitants of an area).
- v. **Harbouring of disease vectors** such as rats and flies, particularly from improperly operated landfills.

Components of a Typical Landfill





Problems

- Construction is costly
- Liners often develop cracks as it react with various chemicals present in the waste
- Rate of decomposition is variable as less oxygen is available due to tight compression of waste

Advantages

- Infectious diseases are minimized
- No air pollution from burning
- Minimal fire hazards
- Methane gas generated is sold as a fuel

INCINERATION

- Incineration is a disposal method in which **solid organic wastes are subjected to combustion so as to convert them into residue and gaseous products.** This method is useful for disposal of residue of both solid waste management and solid residue from waste water management. This process **reduces the volumes of solid waste to 20 -30 % of the original volume.**
- An incinerator is a unit or facility used to burn trash and other types of waste until it is reduced to ash. It is constructed of **heavy, well-insulated materials, so that it does not give off extreme amounts of external heat.** The high levels of heat are kept inside the furnace or unit so that the waste is burned quickly and efficiently. If the heat were allowed to escape, the waste would not burn as completely or as rapidly.
- Incineration and other high temperature waste treatment systems are sometimes described as “**thermal treatment**”. **Incinerators convert waste materials into heat, gas, steam and ash.**
- It is recognized as a **practical method of disposing of certain hazardous waste materials.**

Advantages:

- Recovery of useful materials and energy
- Varying solid waste loads are converted to small volumes of ash which can be controlled

Limitations:

- Controversial method of waste disposal, due to issues such as emission of gaseous pollutants.
- Fly ash left in the furnace contains dangerous toxins which has several adverse impacts on health and environment

COMPOSTING

- Composting is a **biological process in which micro-organisms, mainly fungi and bacteria, convert degradable organic waste into humus like substance.** which looks like soil, is high in carbon and nitrogen and is an excellent medium for growing plants, improves the soil conditions and fertility.
- Due to shortage of space for landfill in bigger cities, the biodegradable yard waste (kept separate from the municipal waste) is allowed to degrade or decompose in a medium.

Optimum conditions for composting

- Tempertaure – 40-50 °C
- Moisture – 40-70%
- Air - 0.5 to 0.8 m³/day/kg
- Carbon: nitrogen ratio - 35 to 50: 1
- Carbon:Phosphorus ration – 100:1



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Vermicomposting

- In Vermi-composting method, **worms are added to the compost**. These help to break the waste and the added excreta of the worms makes the compost very rich in nutrients.

Method

- To make a compost pit, you have to select a cool, shaded corner of the garden or the school compound and dig a pit, which ideally should be 3 feet deep. This depth is convenient for aerobic composting as the compost has to be turned at regular intervals in this process.
- Preferably the pit should be lined with granite or brick to prevent nitrite pollution of the subsoil water, which is known to be highly toxic. Each time organic matter is added to the pit it should be covered with a layer of dried leaves or a thin layer of soil which allows air to enter the pit thereby preventing bad odour.
- At the end of 45 days, the rich pure organic matter is ready to be used.



Benefits of composting

- Clean, cheap, and safe
- Compost allows the soil to retain more plant nutrients over a longer period.
- It supplies part of the 16 essential elements needed by the plants.
- It helps reduce the adverse effects of excessive alkalinity, acidity, or the excessive use of chemical fertilizer.
- It makes soil easier to cultivate.
- It helps keep the soil cool in summer and warm in winter.
- It aids in preventing soil erosion by keeping the soil covered.
- It helps in controlling the growth of weeds in the garden.
- Significantly reduce the amount of disposable garbage.
- Increases the soil's water retention capacity.



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RECYCLING AND REUSE

- **Recycling** is the recovery and **reuse** of materials from wastes.
- **Solid waste** recycling refers to the reuse of manufactured goods from which resources such as steel, **copper**, or **plastics** can be recovered and reused. Recycling and recovery is only one phase of an integrated approach to solid **waste management** that also includes reducing the amount of waste produced prior to **composting**, incinerating, and landfilling.
- Recycling keeps large amounts of solid waste out of landfills, conserve resources, and save energy. The technology of recycling involves collection, separation, preparing the material to buyer's specifications, sale to markets, processing, and the eventual reuse of materials. Separation and collection is only the first step; if the material is not also processed and returned to commerce, then it is not being recycled. In many parts of the country, markets are not yet sufficiently developed to handle the growing supply of collected material.
- The materials recycled today include **aluminum**, paper, glass, plastics, iron and steel, scrap tires, and used oil.



Four R's to be followed to manage the problem of solid waste

- The waste management hierarchy breaks this process down into four parts, also known as the **Four R's: Refuse, Reduce, Reuse, and Recycle**. Most people know them by heart as a slogan, but they're actually a hierarchy that starts with the first and most preferable option, and goes down the ladder. Don't recycle plastic, for example, when you can reuse it, or—even better—quit using it altogether in the first place.

1. Refuse

- Cut down the waste simply by refusing to buy individually packaged items and single-use plastics.
- By bringing metal straws and coffee tumblers for drinks
- Use of eco-bags for shopping instead of using paper and plastic bags
- Opt to receive the bills online instead of paper bills
- Avoid using disposable plates, utensils, and cups



2. Reduce

- Part of throwing away less is buying less, and being more mindful as a consumer. Before you purchase something, consider whether you really need it, and how many uses you'll get out of it.
- Instead of buying cheap clothes that need to be replaced in a few months, invest in well-made items that will last for years
- Take good care of your clothes, shoes, and bags so they don't wear out as quickly
- Donate old clothes and any items in good condition to charity, instead of tossing them out

3. Reuse

- Before tossing out your trash, consider whether it can be repurposed as something else.
- Use a chipped mug as a pen or toothbrush holder
- Re-use old jars, pots, bottles, and plastic tupperware
- Purchase rechargeable batteries
- Compost coffee grounds and tea bags and use them as fertilizer
- Turn single-use plastics into ecobricks



4. Recycle

After refusing, reducing, and reusing, recycling is the last resort in the waste management hierarchy. Make sure to properly sort and clean your items before bringing them to recycling centers.

Advantages

- Less utilization of raw materials
- Reduces environmental impacts arising from waste disposal and treatment
- Saves money, energy required to manufacture new products
- Reduces pollution
- Recycling paper will reduce deforestation
- Reuse of metals will reduce mining and melting of ores
- Saves land space

DUMPING

- An **open dumping** is defined as a **land disposal site at which solid wastes are disposed of in a manner that does not protect the environment, are susceptible to open burning**, and are exposed to the elements, vectors, and scavengers.
- Open dumping can include solid waste disposal facilities or practices that pose a reasonable probability of adverse effects on health or the environment.
- It is the **breeding ground for flies, rats, insects and pathogens**. The rain water from these dumps contaminates nearby land and water thereby spreading diseases
- It is practiced extensively in India as it is cheap.



References

- E. Barucha, *Textbook of Environmental Studies for Undergraduate Courses*, Universities Press (India) Pvt. Ltd., 2005.
- S. Chawla, *A Textbook of Environmental Studies*, McGraw Hill Education Private Limited, 2012
- G. T. Miller, *Environmental Science*, Thomas Learning, 2012
- R. Rajagopalan, *Environmental Studies: From Crisis to Cure*, 2nd Edition, Oxford University Press, 2011.
- A.K. De, *Environmental Chemistry*, New Age Int. Publ. 2012.
- A. Kaushik and C.P. Kaushik, *Perspectives in Environment Studies*, 4th Edition, New Age International Publishers, 2013.
- <https://greencleanguide.com/5-ways-to-safely-dispose-of-your-electronic-waste/>
- <http://www.yourarticlelibrary.com/essay/hazard-waste-treatment-hwt-technologies-physical-and-chemical-treatment-methods/27478>
- <https://www.britannica.com/technology/hazardous-waste-management/Treatment-storage-and-disposal>
- <https://www.conserve-energy-future.com/radioactive-pollution-causes-effects-solutions.php>
- <https://www.envis.org/technology/87-swm/manual-on-municipal-solid-waste-management/267-classification-of-solid-wastes>
- <https://www.britannica.com/technology/solid-waste-management/Solid-waste-collection>
- <http://www.yourarticlelibrary.com/waste-management/solid-waste-management-types-sources-effects-and-methods-of-solid-waste-management/9949>



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- <https://www.encyclopedia.com/environment/encyclopedias-almanacs-transcripts-and-maps/solid-waste-recycling-and-recovery>
- <https://www.esquiremag.ph/culture/lifestyle/how-to-reduce-plastic-waste-four-rs-a00225-20180806>
- https://www.usgs.gov/special-topic/water-science-school/science/dissolved-oxygen-and-water?qt-science_center_objects=0#qt-science_center_objects
- <https://www.arviatechnology.com/what-are-the-three-stages-of-wastewater-treatment/>
- https://en.wikipedia.org/wiki/Sewage_treatment
- <https://www.britannica.com/technology/wastewater-treatment/Primary-treatment>
- https://www.usgs.gov/special-topic/water-science-school/science/biological-oxygen-demand-bod-and-water?qt-science_center_objects=0#qt-science_center_objects
- https://www.researchgate.net/figure/Mechanism-of-Ozone-Layer-Destruction-by-CFC-in-Stratosphere_fig4_228279972
- <https://www.conserve-energy-future.com/causes-and-effects-of-acid-rain.php>
- <https://byjus.com/physics/electrostatic-precipitator/>
- https://energyeducation.ca/encyclopedia/Wet_scrubber
- <https://pubs.usgs.gov/fs/2008/3097/pdf/CarbonFS.pdf>
- <http://www.nec.co.jp/eco/en/annual2006/02/2-1.html>